Poisson Distribution: Step-by-Step Proofs of the Main Properties

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Abstract

Complete, self-contained proofs of the main properties of the Poisson distribution: the probability mass function (as the limit of the binomial law), the CDF, the moment generating function, mean, variance, standard deviation, mode, the reliability and hazard functions (in both the waiting-time and the discrete-count settings), and the normal approximation. This revision corrects the earlier version, in particular the definition of the hazard function and the proofs of the PMF and the mode.

1 Proofs

Throughout, $X \sim \text{Poisson}(\lambda)$ with $\lambda > 0$, and $p_k = P(X = k) = e^{-\lambda}\lambda^k/k!$ for $k = 0, 1, 2, \dots$ We use the exponential series $e^x = \sum_{j\geq 0} x^j/j!$, valid for every real x.

1.1 Probability mass function (Poisson limit theorem)

Theorem 1. *Let $X_n \sim \text{Bin}(n, \lambda/n)$. For every fixed $k \in \{0, 1, 2, \dots\}$,*

$$\lim_{n\to\infty} P(X_n = k) = \frac{e^{-\lambda}\lambda^k}{k!}.$$

Proof.

$$\begin{aligned} P(X_n = k) &= \binom{n}{k}\Big(\frac{\lambda}{n}\Big)^k\Big(1 - \frac{\lambda}{n}\Big)^{n-k} \\ &= \frac{\lambda^k}{k!} \cdot \underbrace{\frac{n(n-1)\cdots(n-k+1)}{n^k}}_{=\prod_{j=0}^{k-1}(1-j/n)} \cdot \Big(1 - \frac{\lambda}{n}\Big)^n \cdot \Big(1 - \frac{\lambda}{n}\Big)^{-k}. \end{aligned}$$

As $n \to \infty$ with k fixed: the product of k factors tends to 1; $(1-\lambda/n)^n \to e^{-\lambda}$; and $(1-\lambda/n)^{-k} \to 1$. Hence $P(X_n = k) \to e^{-\lambda}\lambda^k/k!$. □

Remark 1. (i) The p_k form a probability distribution because $\sum_{k\geq 0} p_k = e^{-\lambda}\sum_{k\geq 0}\lambda^k/k! = e^{-\lambda}e^{\lambda} = 1$. (ii) *Poisson process.* Suppose events in $[0, t]$ occur independently, at most one per small sub-interval, with probability $\lambda t/n$ in each of n equal sub-intervals. Then the count $N(t)$ is $\text{Bin}(n, \lambda t/n)$, and letting $n \to \infty$ gives $N(t) \sim \text{Poisson}(\lambda t)$, i.e. $P(N(t) = k) = e^{-\lambda t}(\lambda t)^k/k!$.

1.2 Cumulative distribution function

By definition

$$F(k) = P(X \le k) = \sum_{i=0}^{k} \frac{e^{-\lambda}\lambda^i}{i!}, \qquad P(X > k) = 1 - F(k) = \sum_{i=k+1}^{\infty} \frac{e^{-\lambda}\lambda^i}{i!}.$$

(The exponential series shows that the two sums are complementary, since the full series sums to 1.)

Proposition 2 (Incomplete-gamma form). $F(k) = \dfrac{1}{k!}\displaystyle\int_{\lambda}^{\infty} s^k e^{-s}\, ds = \dfrac{\Gamma(k+1,\lambda)}{k!}$. *In particular F is strictly decreasing in λ.*

Proof. Differentiate in λ; each term gives $\frac{d}{d\lambda}\frac{e^{-\lambda}\lambda^i}{i!} = -\frac{e^{-\lambda}\lambda^i}{i!} + \frac{e^{-\lambda}\lambda^{i-1}}{(i-1)!}$ (the last term is absent for $i = 0$). Summing over $i = 0, \ldots, k$ the sum telescopes: $\frac{d}{d\lambda}F(k) = -\frac{e^{-\lambda}\lambda^k}{k!} < 0$. Since $F(k) \to 0$ as $\lambda \to \infty$, integrating from λ to ∞ gives $F(k) = \int_{\lambda}^{\infty} \frac{e^{-s}s^k}{k!} ds$. □

1.3 Moment generating function

Theorem 3. $M(t) = E[e^{tX}] = \exp\big(\lambda(e^t - 1)\big)$ *for all real t.*

Proof.

$$M(t) = \sum_{k=0}^{\infty} e^{tk}\frac{e^{-\lambda}\lambda^k}{k!} = e^{-\lambda}\sum_{k=0}^{\infty}\frac{(\lambda e^t)^k}{k!} = e^{-\lambda}e^{\lambda e^t} = e^{\lambda(e^t-1)}.$$ □

1.4 Mean

Theorem 4. $E[X] = \lambda$.

Proof. The $k = 0$ term vanishes, and $k/k! = 1/(k-1)!$ for $k \ge 1$. Substituting $j = k - 1$:

$$E[X] = \sum_{k=1}^{\infty} k\,\frac{e^{-\lambda}\lambda^k}{k!} = e^{-\lambda}\sum_{k=1}^{\infty}\frac{\lambda^k}{(k-1)!} = \lambda e^{-\lambda}\sum_{j=0}^{\infty}\frac{\lambda^j}{j!} = \lambda e^{-\lambda}e^{\lambda} = \lambda.$$ □

1.5 Variance

Theorem 5. $\mathrm{Var}(X) = \lambda$.

Proof. Write $k^2 = k(k-1) + k$. For $k \ge 2$, $k(k-1)/k! = 1/(k-2)!$, so with $j = k - 2$

$$E[X(X-1)] = e^{-\lambda}\sum_{k=2}^{\infty}\frac{\lambda^k}{(k-2)!} = \lambda^2 e^{-\lambda}\sum_{j=0}^{\infty}\frac{\lambda^j}{j!} = \lambda^2.$$

Hence $E[X^2] = E[X(X-1)] + E[X] = \lambda^2 + \lambda$ and $\mathrm{Var}(X) = E[X^2] - (E[X])^2 = \lambda^2 + \lambda - \lambda^2 = \lambda$.

Cross-check via the MGF: $M'(0) = \lambda$ and $M''(0) = \lambda^2 + \lambda$ (differentiate $M(t) = e^{\lambda(e^t-1)}$ twice), giving the same values. □

1.6 Standard deviation

$\sigma = \sqrt{\mathrm{Var}(X)} = \sqrt{\lambda}$. The equality of mean and variance (*equidispersion*) is the property tested by the dispersion diagnostic used in the regression applications of the paper.

1.7 Mode

Theorem 6. *If $\lambda \notin \mathbb{Z}$, the unique mode is $\lfloor \lambda \rfloor$. If $\lambda \in \mathbb{Z}$, the modes are $\lambda - 1$ and λ.*

Proof. For $k \geq 1$ the ratio of consecutive probabilities is

$$\frac{p_k}{p_{k-1}} = \frac{e^{-\lambda}\lambda^k/k!}{e^{-\lambda}\lambda^{k-1}/(k-1)!} = \frac{\lambda}{k}.$$

So $p_k > p_{k-1}$ iff $k < \lambda$, $p_k = p_{k-1}$ iff $k = \lambda$, and $p_k < p_{k-1}$ iff $k > \lambda$. The sequence therefore increases while $k < \lambda$ and decreases once $k > \lambda$. If λ is not an integer, the last increase is at $k = \lfloor \lambda \rfloor$ and the next step is a decrease, so $\lfloor \lambda \rfloor$ is the unique maximum. If λ is an integer, $p_\lambda = p_{\lambda-1}$ and both are maximal (for $\lambda < 1$ the sequence decreases from $k = 0$, so the mode is $0 = \lfloor \lambda \rfloor$). □

1.8 Reliability and hazard functions

Two different objects are called "reliability" in connection with the Poisson law; they must not be mixed.

(A) Waiting time in a Poisson process. Let events occur at rate λ per unit time and let T be the time of the first event. Then $\{T > t\} = \{N(t) = 0\}$, so by the Remark in Section 1.1

$$R(t) = P(T > t) = e^{-\lambda t}, \qquad f(t) = -R'(t) = \lambda e^{-\lambda t}, \qquad Z(t) = \frac{f(t)}{R(t)} = \lambda.$$

So $T \sim \mathrm{Exp}(\lambda)$ and the hazard is the constant λ (the memoryless property). This is the reliability function used in reliability engineering.

(B) The count as a discrete "lifetime". Using the index k itself as the variable, define the survival function and the *discrete hazard*

$$S(k) = P(X > k) = 1 - F(k),$$
$$h(k) = P(X = k \mid X \geq k) = \frac{P(X = k)}{P(X \geq k)} = \frac{p_k}{1 - F(k-1)}, \qquad F(-1) := 0.$$

Note that the denominator is $P(X \geq k)$, not $P(X > k)$: a hazard is a *conditional probability*, so it lies in $(0, 1]$. (The ratio $p_k/P(X > k)$ is not a probability and can exceed 1; for $\lambda = 5$ it reaches about 3.06 at $k = 19$.)

Proposition 7. *$h(0) = e^{-\lambda}$; $0 < h(k) < 1$ for every k; h is strictly increasing in k; and $h(k) \to 1$ as $k \to \infty$.*

Proof. $h(0) = p_0/P(X \geq 0) = e^{-\lambda}$. Since $P(X \geq k) = p_k + P(X > k)$ and $P(X > k) > 0$, we get $h(k) < 1$. Next,

$$\frac{1}{h(k)} = \sum_{j=0}^{\infty} \frac{p_{k+j}}{p_k} = \sum_{j=0}^{\infty} \frac{\lambda^j\, k!}{(k+j)!} = 1 + \sum_{j=1}^{\infty} \frac{\lambda^j}{(k+1)(k+2)\cdots(k+j)}.$$

For each $j \geq 1$ the term decreases strictly in k, so $1/h(k)$ strictly decreases, i.e. h strictly increases. Each term is also bounded by $\lambda^j/j!$ (summable) and tends to 0 as $k \to \infty$, so by dominated convergence $1/h(k) \to 1$. □

Remark 2. The *increasing* hazard in (B) is a statement about how the conditional chance of "stopping at k" grows with k; it is a property of the count distribution, not of waiting time. Contrast it with the constant hazard λ in (A).

1.9 Normal approximation for large λ

Theorem 8. *If $Z_\lambda = (X-\lambda)/\sqrt{\lambda}$, then $Z_\lambda \Rightarrow N(0,1)$ as $\lambda \to \infty$. Hence $X \approx N(\mu=\lambda,\ \sigma^2=\lambda)$.*

Proof. With $u = s/\sqrt{\lambda}$, the MGF of Z_λ is

$$M_{Z_\lambda}(s) = e^{-s\sqrt{\lambda}}\, e^{\lambda(e^u-1)} = \exp\!\big(\lambda(e^u - 1 - u)\big),$$

and $e^u - 1 - u = u^2/2 + O(u^3)$, so $\lambda(e^u-1-u) = s^2/2 + O(\lambda^{-1/2}) \to s^2/2$. Thus $M_{Z_\lambda}(s) \to e^{s^2/2}$, the MGF of $N(0,1)$, and the claim follows from Lévy's continuity theorem. $\square$

In practice a continuity correction improves the approximation: $P(X \le k) \approx \Phi\big((k + \tfrac{1}{2} - \lambda)/\sqrt{\lambda}\big)$.